

SMC Premium/Discount Confluence Engine — v17.22 Completion Report

Summary

v17.22 — Gate EQ Line to Active Zone Only (Visual-Only Update)

Problem

The EQ (Equilibrium) line was rendering across ALL historical bars via `plot()` , creating a visible squiggle/wave effect as `h4RangeHigh` and `h4RangeLow` values changed over time. The P&D zone boxes only render on the last 50 bars (`bar_index - 50` to `bar_index + 500`), but the EQ line was plotted everywhere `h4RangeHigh` / `h4RangeLow` had values.

Solution

Gated the `eq_level` value with `bar_index >= last_bar_index - 50` so the EQ line only renders within the active P&D zone region, matching the box's left edge extent exactly.

Delivery Checklist

#	Task	Status
1	Backup v17.21 → smc_premium_discount_indica tor_v17.21_backup.pine	✓
2	Identified zone active gating: bar_index >= last_bar_index - 50	✓
3	Implemented active_eq variable with zone gating	✓
4	Updated plot() call to use active_eq instead of eq_level	✓
5	Version incremented: v17.21 → v17.22 (40 references)	✓
6	Title string updated	✓
7	HUD header version updated	✓
8	All 4 alertcondition JSON payloads updated	✓
9	All comments updated	✓
10	ZERO v17.21 references re- main in v17.22 file	✓
11	Plot budget unchanged: 34/64	✓
12	ZERO trading logic changes	✓
13	Branch pushed: feat/v17.22- eq-gate-active-zone	✓
14	PR #6 created	✓

Implementation Detail

Zone Active Gating Variable

The P&D zones don't use a traditional boolean `zoneActive` variable. Instead, the zone boxes are drawn conditionally with:

```
if showH4Zones and barstate.islast
    pdPremiumBox := box.new(bar_index - 50, ..., bar_index + 500, ...)
```

The boxes extend from `bar_index - 50` (left) to `bar_index + 500` (right). The EQ line should only be visible in this same region.

EQ Gating Code (v17.22)

```
// EQ Line - 50% midpoint of P&D Zone
eq_level = na(h4RangeHigh) or na(h4RangeLow) ? na : (h4RangeHigh + h4RangeLow) / 2

// v17.22: Only show EQ when within active zone region (last 50 bars matches P&D box extent)
active_eq = (bar_index >= last_bar_index - 50) ? eq_level : na

plot(showH4Zones and showEquilibriumLine ? active_eq : na, title="EQ", color=color.yelow, linewidth=3, style=plot.style_line)
```

Before vs. After

Property	v17.21	v17.22
EQ Visibility	ALL historical bars	Last 50 bars only
Squiggle Effect	Present (values change per bar)	Eliminated
Zone Alignment	Mismatched (EQ everywhere, boxes last 50 bars)	Matched perfectly
Plot Budget	34/64	34/64 (unchanged)

Files Modified

- `smc_premium_discount_indicator_v17.22.pine` — New production script
- `smc_premium_discount_indicator_v17.21_backup.pine` — Backup of previous version

GitHub

- **Branch:** `feat/v17.22-eq-gate-active-zone`
- **PR:** [#6](https://github.com/Edal59/smc-tracker-railway/pull/6) (<https://github.com/Edal59/smc-tracker-railway/pull/6>)

Expected Behavior After Installation

- EQ line (yellow, width 3) appears **only** within the P&D zone region
- No squiggle or wave across historical bars
- EQ line aligns perfectly with Premium (red) and Discount (green) zone boxes
- Line appears as a flat horizontal at the exact 50% midpoint of the active zone

ZERO Logic Changes Confirmed

-  Guardian filter: unchanged
-  Confluence scoring: unchanged
-  Signal generation: unchanged
-  Alert payloads: unchanged (version string only)
-  HUD layout: unchanged (version display only)
-  Zone boxes: unchanged
-  S&D zones: unchanged